

RAHMANI

NASR-EDDINE

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Electromechanical Engineer

Electrical & Industrial Systems | Control Systems & Automation



EDUCATION

- 2015-2018 *State Engineering Degree in Electromechanical Engineering — ENSMR (formerly ENIM), Rabat*
- 2012-2015 *Preparatory Classes for Engineering Schools (CPGE) — Mathematics & Physics (MP)*

PROFESSIONAL EXPERIENCE

- Jan 2023 - Present **Electromechanical Systems Engineer — Vanguard Center / UM6P, Rabat Technopolis**
 - Design and integration of complex electromechanical systems from architecture through validation.
 - Development and validation of control architectures for electric actuator systems.
 - System commissioning, field testing, troubleshooting, and performance optimization under real-world operating conditions.
 - Technical coordination, including planning, milestones, and system integration activities.
- Jan 2020 - Dec 2022 **Control Systems Engineer — UM6P, Benguerir**
 - Developed and implemented real-time control algorithms (PID, cascaded control, adaptive control, MPC) for autonomous ground and aerial systems.
 - Integrated LiDAR, RTK-GPS, and IMU sensors for navigation and closed-loop control.
 - Performed HIL/SIL validation and field testing under real-world operating conditions.
- Dec 2018 - Dec 2019 **Industrial Maintenance Engineer — AIC / Delta Holding Group, Skhirat**
 - Supervised preventive and corrective maintenance of production equipment, including laser cutting machines, CNC systems, press brakes, and powder coating lines.
 - Troubleshoot electrical and mechanical failures to improve equipment availability and maintain production continuity.
 - Contributed to plant expansion projects and the development of new products (R&D).
- Mar - Jun 2018 **Electrical Engineering Intern — NOVEC, Rabat Technopolis**
 - Assessed the facility’s MV/LV electrical distribution network and performed load analysis.
 - Designed and sized a 200 kWp grid-connected photovoltaic system for on-site self-consumption.
- Jul - Sep 2017 **Maintenance & Reliability Engineering Intern — ONCF Group, Oujda**
 - Performed FMEA on DH/DK locomotives to identify critical electromechanical components and support preventive maintenance improvements.

TECHNICAL SKILLS

- Engineering
 - Control Systems & Automation:** PID, MPC/NMPC, PLC programming (LAD/FBD/SCL), HMI, Siemens TIA Portal, process control.
 - Simulation & Programming:** MATLAB/Simulink, HIL/SIL, Python, C/C++.
 - Electrical & Electromechanical Systems:** LV power distribution, electrical machines, power electronics, motor protection.
 - Commissioning & Reliability:** field testing, troubleshooting, calibration, preventive/corrective maintenance, FMEA.
 - Industrial Communications & Engineering Tools:** Modbus, CAN, AutoCAD, Caneco BT, PVsyst.
- Languages
 - French:** Professional proficiency. **English:** Professional proficiency.